Assignment 16: Systems of Equations
DO NOT WRITE ON THIS PAPER

Part A

Consider the following system of equations.

$$x + 2y = 5$$
$$3x - 2y = -1$$

Solve the system of equations using substitution or elimination. Show your work and the check step.

Part B

Consider the following systems of equations:

$$y - x = 1$$
$$2y - 2x = 2$$

Use substitution or elimination to solve the system of equations. How many solutions does the system of equations have?

Part C

Consider the following systems of equations:

$$y = x + 1$$
$$y - x = 2$$

Use substitution or elimination to solve the system of equations. How many solutions does the system of equations have?

Question 1

Big Question: What does it mean to solve a system of equations?

Use the following words in your response: *equal*, *variables*, *intersection*

Then answer the following word problem:

Juan will rent a car for the weekend. He can choose one of two plans. The first plan has an initial fee of $45 and costs an additional $0.20 per mile driven. The second plan has no initial fee but costs $0.70 per mile driven.

For what amount of driving do the two plans cost the same? What is the cost when the two plans cost the same?

Question 2 Work	Question 2 Answer
Solve the following system of equations. Show your work and the check step. $$\begin{gathered} 3y+9=x \\ 3x+2y=5 \end{gathered}$$	Write your answer in your Class Notes

Question 3 Work	Question 3 Answer
Melissa is a software saleswoman. Her monthly salary is $2,200 plus an additional $40 for every copy of History is Fun she sells. Let S represent her total salary this month (in dollars) and let N represent the number of copies of History is Fun she sells. Write an equation relating S to N. Then, use this equation to find how many copies of History is Fun she needs to sell to earn a total salary of $6,520 this month.	Write your answer in your Class Notes

Additional Space